Temperature Scales

Answer key with worked solutions

A. Understanding

1. The factor 9/5.

> The same temperature interval is 180 °F or 100 °C, so 1 °C is the same size as 180/100 = **9/5 = 1.8 Fahrenheit degrees**. To convert, multiply the Celsius value by 9 °F/5 °C (and allow for the different zero points with the 32 °F).

2. Kelvin vs Celsius.

> Any two of: kelvin has **no degree sign** (300 K, not 300 °K); kelvin **starts at absolute zero**, so it is never negative; kelvin must be used in gas-law and other calculations involving ratios of temperatures. (The size of one kelvin and one Celsius degree is the same.)

B. Celsius and Fahrenheit

3. To °F.

> $T_\mathrm{F} = (9\ ^\circ\mathrm{F}/5\ ^\circ\mathrm{C}) \times T_\mathrm{C} + 32\ ^\circ\mathrm{F}$
>
> (a) $(9\ ^\circ\mathrm{F}/5\ ^\circ\mathrm{C})(100\ ^\circ\mathrm{C}) + 32\ ^\circ\mathrm{F} = 180\ ^\circ\mathrm{F} + 32\ ^\circ\mathrm{F} = \mathbf{212.0\ ^\circ F}$
>
> (b) $(9\ ^\circ\mathrm{F}/5\ ^\circ\mathrm{C})(-10\ ^\circ\mathrm{C}) + 32\ ^\circ\mathrm{F} = -18\ ^\circ\mathrm{F} + 32\ ^\circ\mathrm{F} = \mathbf{14.0\ ^\circ F}$
>
> (c) $(9\ ^\circ\mathrm{F}/5\ ^\circ\mathrm{C})(37.5\ ^\circ\mathrm{C}) + 32\ ^\circ\mathrm{F} = 67.5\ ^\circ\mathrm{F} + 32\ ^\circ\mathrm{F} = \mathbf{99.5\ ^\circ F}$

4. To °C.

> $T_\mathrm{C} = (5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F}) \times (T_\mathrm{F} - 32\ ^\circ\mathrm{F})$
>
> (a) $(5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(50\ ^\circ\mathrm{F} - 32\ ^\circ\mathrm{F}) = (5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(18\ ^\circ\mathrm{F}) = \mathbf{10.0\ ^\circ C}$
>
> (b) $(5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(0\ ^\circ\mathrm{F} - 32\ ^\circ\mathrm{F}) = (5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(-32\ ^\circ\mathrm{F}) = \mathbf{-17.8\ ^\circ C}$
>
> (c) $(5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(451\ ^\circ\mathrm{F} - 32\ ^\circ\mathrm{F}) = (5\ ^\circ\mathrm{C}/9\ ^\circ\mathrm{F})(419\ ^\circ\mathrm{F}) = \mathbf{232.8\ ^\circ C}$

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C. Kelvin

5. To kelvin.

> (a) $25\ ^\circ\text{C} + 273.15 = \mathbf{298.15\ K}$
>
> (b) $-196\ ^\circ\text{C} + 273.15 = \mathbf{77.15\ K}$ (about 77 K)
>
> (c) First to °C: $(5\ ^\circ\text{C}/9\ ^\circ\text{F})(212\ ^\circ\text{F} - 32\ ^\circ\text{F}) = 100\ ^\circ\text{C}$; then $100\ ^\circ\text{C} + 273.15 = \mathbf{373.15\ K}$

6. 300 K.

> $T_\text{C} = 300\ \text{K} - 273.15 = \mathbf{26.85\ ^\circ C}$ (26.8 °C to three significant figures: an exact 5 rounds to the even digit)
>
> $T_\text{F} = (9\ ^\circ\text{F}/5\ ^\circ\text{C})(26.85\ ^\circ\text{C}) + 32\ ^\circ\text{F} = 48.33\ ^\circ\text{F} + 32\ ^\circ\text{F} = \mathbf{80.3\ ^\circ F}$

D. Temperature changes

7. A rise of 15 °C.

> A kelvin is the same size as a Celsius degree, so the rise is **15 K**. In Fahrenheit, use only the size factor (the 32 °F offsets cancel for a change):
>
> $(1.8\ ^\circ\text{F}/^\circ\text{C})(15\ ^\circ\text{C}) = \mathbf{27\ ^\circ F}$

E. Where the scales meet

8. −40.

> Let the same number be x. Then $x = \frac{9}{5}x + 32$, so $x - \frac{9}{5}x = 32$, $-\frac{4}{5}x = 32$ and $x = \mathbf{-40}$.
>
> Check: $(9\ ^\circ\text{F}/5\ ^\circ\text{C})(-40\ ^\circ\text{C}) + 32\ ^\circ\text{F} = -72\ ^\circ\text{F} + 32\ ^\circ\text{F} = -40\ ^\circ\text{F}$ ✓

9. Kelvin and Fahrenheit.

> Write °F in terms of K: $T_\text{F} = \frac{9}{5}(T_\text{K} - 273.15) + 32$. Set $T_\text{F} = T_\text{K} = x$:
>
> $x = \frac{9}{5}x - 491.67 + 32$, so $-\frac{4}{5}x = -459.67$ and $x = \mathbf{574.59}$
>
> So **574.59 K = 574.59 °F**, which is $574.59\ \text{K} - 273.15 = \mathbf{301.44\ ^\circ C}$.
>
> Kelvin and Celsius numbers are **never** equal: $T_\text{K} = T_\text{C} + 273.15$ always differ by 273.15, because the two scales have the same size of degree but different zero points.

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